# Siyang Li

*Brinson Prize Postdoctoral Fellow, University of California, Berkeley*

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## RESEARCH

- Distances to nearby galaxies and the Hubble constant with three independent distance indicators: Cepheids, the tip of the red giant branch (TRGB), and the J-region asymptotic giant branch (JAGB).
- PI of the CHASE spectral survey, to standardize the JAGB as a distance indicator independent of Cepheids & TRGB.
- Lead organizer, CosmoVerse Data Challenge & School.

## APPOINTMENTS

**Brinson Prize Postdoctoral Fellow**, UC Berkeley (Host: Alexei V. Filippenko) 2025–present

## FELLOWSHIPS

**Brinson Prize Postdoctoral Fellowship** ($360,000) 2025–2028

**International Space Science Institute (ISSI) Postdoctoral Fellowship** (declined) 2025

**National Science Foundation Graduate Research Fellowship** ($108,000) 2022–2025

## TELESCOPE TIME

**PI, Lick Observatory Shane 3-meter Telescope** — 3 nights each semester 2026A, 2026B
*Refining a New Hubble Tension Probe: Testing J-region Asymptotic Giant Branch Homogeneity Assumptions*

**PI, Apache Point Observatory ARC 3.5-meter Telescope** — 7 half-nights 2024–2025
*Towards Standardization of the J-region Asymptotic Giant Branch: Investigating Asymmetric Luminosity Functions with Carbon Star Spectra*

**PI, Hubble Space Telescope Cycle 32 Bridge**, GO-17911 — 8 orbits ($25,000) 2024
*Building the Last Bridge Between the PHAT & PHATTER Surveys and Carbon Star Cosmology*

**Co-I, Hubble Space Telescope Cycle 31**, GO-17520 (PI: L. Breuval) — 42 orbits 2023
*A 1% cross-calibration of Cepheids, TRGB, and JAGB in five nearby galaxies with HST*

## EDUCATION

**Johns Hopkins University** 2020–2025
*Doctor of Philosophy in Astronomy & Astrophysics*
*Master of Arts in Physics* 2020–2023
Dissertation: *Testing Hubble Tension Systematics using Cepheids, Tip of the Red Giant Branch, and J-region Asymptotic Giant Branch*
Thesis Advisor: Professor Adam G. Riess

**University of California, Berkeley** 2016–2020
*Bachelor of Arts, High Honors in Physics, cum laude*
Honors Thesis: *Characterization of a High Efficiency Silicon Photomultiplier for Astrophysical Transient Searches*
Thesis Advisor: Professor George F. Smoot III

## FIRST-AUTHOR PUBLICATIONS

*Citations to all 13 first-author publications (8 refereed, 1 submitted, 4 proceedings) as of October 16, 2026: 248.*
*†Student mentored by S. Li.*

9. **Siyang Li**, Alexei V. Filippenko, Arief Ahmad, Adam G. Riess, Stefano Casertano, Thomas G. Brink, Richard I. Anderson, Caroline D. Huang, Louise Breuval, Noriyuki Matsunaga, Gracelynn Jost†, Yash Mehta†, Elaina Sadler†, Julian Shapiro†. "CHASE Spectral Survey I: Measuring Contamination in the Color-selected J-region Asymptotic Giant Branch Distance Indicator." *Submitted to ApJ* (2026). ADS
8. **Siyang Li**, Adam G. Riess, Gagandeep S. Anand, Daniel Scolnic, Yukei S. Murakami, Dillon Brout, Erik R. Peterson. "The Complete Sample of Available SNe Ia Luminosity Calibrations from the TRGB Observed with either *HST* or *JWST*." *ApJ*, 997, 115 (2026). ADS
7. **Siyang Li**, Adam G. Riess, Daniel Scolnic, Stefano Casertano, Gagandeep S. Anand. "JAGB 2.0: Improved Constraints on the J-region Asymptotic Giant Branch-based Hubble Constant from an Expanded Sample of *JWST* Observations." *ApJ*, 988, 97 (2025). ADS
6. **Siyang Li**, Gagandeep S. Anand, Adam G. Riess, Stefano Casertano, Wenlong Yuan, Louise Breuval, Lucas M. Macri, Daniel Scolnic, Rachael Beaton, Richard I. Anderson. "Tip of the Red Giant Branch Distances with *JWST*. II. I-band Measurements in a Sample of Hosts of 10 Type Ia Supernova Match *HST* Cepheids." *ApJ*, 976, 177 (2024). ADS
5. **Siyang Li**, Adam G. Riess, Stefano Casertano, Gagandeep S. Anand, Daniel Scolnic, Wenlong Yuan, Louise Breuval, Caroline D. Huang. "Reconnaissance with *JWST* of the J-region Asymptotic Giant Branch in Distance Ladder Galaxies: From Irregular Luminosity Functions to Approximation of the Hubble Constant." *ApJ*, 966, 20 (2024). ADS
4. **Siyang Li**, Adam G. Riess, Daniel Scolnic, Gagandeep S. Anand, Jiaxi Wu, Stefano Casertano, Wenlong Yuan, Rachael Beaton, Richard I. Anderson. "Standardized Luminosity of the Tip of the Red Giant Branch utilizing Multiple Fields in NGC 4258 and the CATs Algorithm." *ApJ*, 956, 32 (2023). ADS
3. **Siyang Li**, Stefano Casertano, Adam G. Riess. "A *Gaia* Data Release 3 View on the Tip of the Red Giant Branch Luminosity." *ApJ*, 950, 83 (2023). ADS
2. **Siyang Li**, Stefano Casertano, Adam G. Riess. "A Maximum Likelihood Calibration of the Tip of the Red Giant Branch Luminosity from High Latitude Field Giants using *Gaia* Early Data Release 3 Parallaxes." *ApJ*, 939, 96 (2022). ADS
1. **Siyang Li**, Adam G. Riess, Michael P. Busch, Stefano Casertano, Lucas M. Macri, Wenlong Yuan. "A Sub-2% Distance to M31 from Photometrically Homogeneous Near-infrared Cepheid Period-Luminosity Relations Measured with the *Hubble Space Telescope*." *ApJ*, 920, 84 (2021). ADS

## FIRST-AUTHOR CONFERENCE PROCEEDINGS

4. **Siyang Li**, George F. Smoot III. "Characterization of a high efficiency silicon photomultiplier for millisecond to sub-microsecond astrophysical transient searches." *SPIE Proc.*, 11454, 1145422 (2020). ADS
3. **Siyang Li**, George F. Smoot III, Bruce Grossan, Albert Wai Kit Lau, Marzhan Bekbalanova, Mehdi Shafiee, Thorsten Stezelberger. "Program objectives and specifications for the Ultra-Fast Astronomy observatory." *SPIE Proc.*, 11341, 113411Y (2019). ADS
2. **Siyang Li**, George F. Smoot III. "Characterization of a silicon photomultiplier for the Ultra-Fast Astronomy telescope." *SPIE Proc.*, 11115, 111150A (2019). ADS
1. **Siyang Li**, Jérôme Maire, Maren Cosens, Shelley A. Wright. "Detector characterization of a near-infrared discrete avalanche photodiode 5×5 array for astrophysical observations." *SPIE Proc.*, 11002, 110022G (2019). ADS

## CO-AUTHOR PUBLICATIONS

*Total citations as of October 16, 2026: 1052 (531 excluding the CosmoVerse White Paper)*

10. Stefano Casertano & the $H_0$ Distance Network Collaboration including **Siyang Li**. "The Local Distance Network: A community consensus report on the measurement of the Hubble constant at ~1% precision." *A&A*, 708, A166 (2026). ADS
9. Adam G. Riess, **Siyang Li**, Gagandeep S. Anand, Wenlong Yuan, Louise Breuval, Stefano Casertano, Lucas M. Macri, Daniel Scolnic, Yukei S. Murakami, Alexei V. Filippenko, Thomas G. Brink. "The Perfect Host: *JWST* Cepheid Observations in a Background-free Type Ia Supernova Host Confirm No Bias in Hubble-constant Measurements." *ApJL*, 992, L34 (2025). ADS
8. Eleonora Di Valentino et al. including **Siyang Li**. "The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics." *PDU*, 49, 101965 (2025). ADS
   - *J-region Asymptotic Giant Branch* — Principal Author: **Siyang Li**; Co-authors: A. Riess, S. Casertano, G. Sloan, L. Breuval, B. Zgirski, G. Anand, R. Anderson, C. Huang, D. Scolnic.
   - *Tip of the Red Giant Branch* — Principal Author: R. I. Anderson; Co-authors: G. S. Anand, I. Saltas, G. De Somma, **S. Li**, V. Vansevicius, L. Breuval, A. Riess.
   - *Cepheids* — Principal Author: L. Breuval; Co-authors: R. I. Anderson, G. De Somma, L. Galbany, **S. Li**, L. Macri, L. Perivolaropoulos, A. Riess, V. Vansevicius.
7. Rolf-Peter Kudritzki, Miguel A. Urbaneja, Fabio Bresolin, Lucas M. Macri, Wenlong Yuan, **Siyang Li**, Gagandeep S. Anand, Adam G. Riess. "The Hubble Constant Anchor Galaxy NGC 4258: Metallicity and Distance from Blue Supergiants." *ApJ*, 977, 217 (2024). ADS
6. Adam G. Riess, Daniel Scolnic, Gagandeep S. Anand, Louise Breuval, Stefano Casertano, Lucas M. Macri, **Siyang Li**, et al. "*JWST* Validates *HST* Distance Measurements: Selection of Supernova Subsample Explains Differences in *JWST* Estimates of Local $H_0$." *ApJ*, 977, 120 (2024). ADS
5. Gagandeep S. Anand, Adam G. Riess, Wenlong Yuan, Rachael Beaton, Stefano Casertano, **Siyang Li**, et al. "Tip of the Red Giant Branch Distances with *JWST*: An Absolute Calibration in NGC 4258 and First Applications to Type Ia Supernova Hosts." *ApJ*, 966, 89 (2024). ADS
4. Daniel Scolnic, Adam G. Riess, Jiaxi Wu, **Siyang Li**, Gagandeep S. Anand, Rachael Beaton, Stefano Casertano, Richard I. Anderson, Suhail Dhawan, Xinwei Ke. "CATS: The Hubble Constant from Standardized TRGB and Type Ia Supernova Measurements." *ApJL*, 954, L31 (2023). ADS
3. Louise Breuval, Adam G. Riess, Lucas M. Macri, **Siyang Li**, Wenlong Yuan, Stefano Casertano, Tarini Konchady, Boris Trahin, Meredith J. Durbin, Benjamin F. Williams. "A 1.3% Distance to M33 from *Hubble Space Telescope* Cepheid Photometry." *ApJ*, 951, 118 (2023). ADS
2. Jiaxi Wu, Daniel Scolnic, Adam G. Riess, Gagandeep S. Anand, Rachael Beaton, Stefano Casertano, Xinwei Ke, **Siyang Li**. "Comparative Analysis of TRGBs (CATs) from Unsupervised, Multi-Halo-Field Measurements: Contrast is Key." *ApJ*, 954, 87 (2023). ADS
1. Albert W. K. Lau, Mehdi Shafiee, George F. Smoot III, Bruce Grossan, **Siyang Li**, Zhanat Maksut. "On-sky silicon photomultiplier detector performance measurements for millisecond to sub-microsecond optical source variability studies." *JATIS*, 6(4), 046002 (2020). ADS

## BOOK CHAPTER AND PATENT

### INVITED BOOK CHAPTER

**Siyang Li**, Rachael L. Beaton. "Tip of the Red Giant Branch." Invited chapter in *The Hubble Constant Tension* (Eds. E. Di Valentino & D. Brout), Springer Singapore, 2024. DOI: 10.1007/978-981-99-0177-7_8.

### PATENT

**Siyang Li**, George F. Smoot III, Thorsten Stezelberger. "Silicon Photomultiplier Telescope Camera." U.S. Provisional Patent Application No. 62/940,250, filed November 26, 2019 (provisional only; no full application filed).

## HONORS AND AWARDS

### HONORS

**American Astronomical Society Rodger Doxsey Travel Prize** 2024
*One of 8 recipients selected from 47 entries; funded travel to AAS 245*

**Lindau Nobel Laureate Meeting (Physics), selected participant** 2024
*73rd meeting, Lindau, Germany*

### UNDERGRADUATE AWARDS

**UC Berkeley Regents' and Chancellor's Scholarship** 2016–2020
*Awarded to the top 2% of undergraduates based on academic and extracurricular achievements*

**Hong Kong University of Science and Technology International Research Fellowship** 2017–2019
*Fully covers airfare, accommodation, and meals for research visits to HKUST*

**SPIE Optics and Photonics Education Scholarship** 2019
*One of 84 awarded worldwide, for potential contribution to optics and photonics*

**Other undergraduate research and travel awards**: UC Berkeley Summer Undergraduate Research Fellowship, Physics Undergraduate Research Scholarship, Regents' and Chancellor's Research Fellowship, and Academic Opportunity Fund Grant; Measurement Science Conference Scholarship; Best Poster Award, 8th Applied Optics and Photonics China 2017–2020

## TALKS

### INVITED TALKS

*Galileo Galilei Institute Exploring New Frontiers in Cosmology*, Florence, Italy (virtual) Jul 2026

*Cross-Survey Cosmography: Precision Cosmology, Emerging Tensions and New Physics Conference*, New Delhi, India Apr 2026

*33rd Texas Symposium on Relativistic Astrophysics* (memorial of G. F. Smoot III), Tempe, AZ Dec 2025

*Brinson Prize Fellowship Workshop*, Chicago, IL Nov 2025

*Tensions in Cosmology Conference*, Corfu, Greece Sep 2025

*National Astronomical Observatory of Japan Cosmology Seminar, jointly with La Sapienza*, Tokyo, Japan Jul 2025

*University of Tokyo Astronomy Department Colloquium*, Tokyo, Japan Jul 2025

*Tohoku University Astronomy Department Colloquium*, Sendai, Japan Jun 2025

*Sauverny Observatory (EPFL) Tea Time Talk*, Versoix, Switzerland May 2025

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| *CosmoVerse Workshop@Naples*, Naples, Italy | May 2025 |
| *ISSI Workshop: What's under the H0od? Towards Consensus on the Local Hubble Constant*, Bern, Switzerland | Mar 2025 |
| *University of California, Davis Physics Department Seminar*, Davis, CA | Jan 2025 |
| *Yale University Cosmology Seminar*, New Haven, CT | Nov 2024 |
| *CosmoVerse Seminar* (virtual) | May 2024 |
| *Space Telescope Science Institute JWST Science Conversation*, Baltimore, MD | Mar 2024 |
| *ISSI Annual Meeting: The Stellar Path to the $H_0$ Tension in the Gaia, TESS, LSST and JWST Era*, Bern, Switzerland (virtual) | Jun 2023 |

## CONTRIBUTED TALKS

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| *JASMINE Consortium Workshop: Dynamics and Evolution of the Galactic Center*, Mitaka, Japan | Apr 2026 |
| *sys2025: Systematic and Measurement Errors across the Sciences*, Huntsville, AL | Nov 2025 |
| *JASMINE Consortium Meeting*, Sendai, Japan | Aug 2025 |
| *Kyoto University Department of Astronomy Seminar*, Kyoto, Japan | Jul 2025 |
| *Kavli IPMU APEC Seminar*, Tokyo, Japan | Jul 2025 |
| *American Astronomical Society Meeting 245 Cosmology I*, National Harbor, MD | Jan 2025 |
| *Cosmo'24*, Kyoto, Japan | Oct 2024 |
| *Resolved Stellar Populations 2024 (Celebration of Peter Stetson's Career)*, Florence, Italy | Oct 2024 |
| *CosmoVerse@Kraków 2024*, Kraków, Poland | Jul 2024 |
| *The 6th Neighborhood Workshop at Penn State University*, State College, PA | Apr 2024 |
| *American Astronomical Society Meeting 243 Cosmic Distance Scale II*, New Orleans, LA | Jan 2024 |
| *The 5th Neighborhood Workshop at Penn State University*, State College, PA | Apr 2023 |
| *American Astronomical Society Meeting 241 Cosmology II*, Seattle, WA | Jan 2023 |
| *iid2022 Workshop: Statistical Methods for Event Data*, Guntersville, AL | Nov 2022 |
| *American Astronomical Society Meeting 240 Cosmology III*, Pasadena, CA | Jun 2022 |
| *SPIE Astronomical Telescopes + Instrumentation, Conference 11454* (virtual) | Dec 2020 |
| *SPIE Optics + Photonics, Conference 11115*, San Diego, CA | Aug 2019 |
| *Exploring the Energetic Universe Conference*, Nur-Sultan, Kazakhstan | Jun 2019 |
| Plus internal seminars at Johns Hopkins University and UC Berkeley | 2019–2025 |

## POSTERS

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| *Cambridge Workshops on Cool Stars, Stellar Systems and the Sun (Cool Stars 23)*, Tokyo Bay Area, Japan | Jun 2026 |
| *Statistical Challenges in Modern Astronomy VIII*, State College, PA | Jun 2023 |
| *8th Applied Optics and Photonics China, Conference 10*, Beijing, China | Jul 2019 |
| *SPIE Defense + Commercial Sensing, Conference 11002*, Baltimore, MD | Apr 2019 |

## TEACHING

**Lecturer**, CosmoVerse Data Challenge & School (77 registrants, 31 countries): 5 TRGB lectures — Summer 2026

**Teaching Assistant**, Stars and the Universe: Cosmic Evolution, Johns Hopkins University — Spring 2021, 2022

**Teaching Assistant**, General Physics: Physical Science Majors II, Johns Hopkins University — Fall 2020

**Teaching Assistant**, General Physics Laboratory I, Johns Hopkins University — Fall 2020

## MENTORSHIP

**Gracelynn Jost**, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) — 2026
*Systematics among JAGB measurement variants in the LMC; CosmoVerse Data Challenge*

**Yash Mehta**, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) — 2026
*Field-to-field variation and standardization of the JAGB in M31; CosmoVerse Data Challenge*

**Elaina Sadler**, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) — 2026
*Where the JAGB breaks down, by galaxy type and region morphology; CosmoVerse Data Challenge*

**Julian Shapiro**, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) — 2026
*Planetary nebula luminosity function distance to M101*

**Gurnoor Singh Sandhu**, undergraduate student, UC Berkeley — 2026
*Alternative methods for fitting the TRGB; CosmoVerse Data Challenge*

**Dhiraj Kuniyal**, PhD student, University of Delhi — 2026
*Maximum-likelihood TRGB variants analysis*

**Ryan Lessing**, undergraduate student, Johns Hopkins University — 2023–2024
*Academic and career mentoring through the Physics & Astronomy department program*

## SERVICE

**Grant Proposal Referee**, PRELUDIUM program, National Science Centre (NCN), Poland — 2026

**Lead Organizer**, CosmoVerse Data Challenge & School — 2025–2026

**Member**, UC Berkeley Astronomy Department Spring Colloquium Committee — 2025

**Referee**, *Scientific Reports* (Springer Nature) — 2025

**Member**, Johns Hopkins Physics & Astronomy Mentorship Committee — 2024–2025

**Referee**, *The Astrophysical Journal Supplement Series* — 2024

## OUTREACH

**Public Talk on the Hubble Tension**, Bulgarian Academy of Sciences — 2026
*Outreach talk for the general public in Sofia, Bulgaria*

**Astro Night Public Outreach Talk**, UC Berkeley Department of Astronomy — 2025
*Talk for high school students, community members, and Sequoia Community College students*

**Johns Hopkins Physics Fair Volunteer** — 2025
*Live physics demonstrations to make scientific concepts accessible to the public*

**Johns Hopkins Physics Department Outreach Volunteer** — 2025
*Visited Towson High School to present physics demonstrations and discuss careers in astronomy*

**Program for Advancing the Health Sciences (PATHS) Mentor** — 2021–2022
*Study habits workshop and one-on-one guidance for an Alexandria City public school student*

**Application Resources for College and Higher-Education (ARCH) Mentor and Panelist** — 2021
*Scholarship and essay writing panelist; one-on-one mentor for an underprivileged college applicant*